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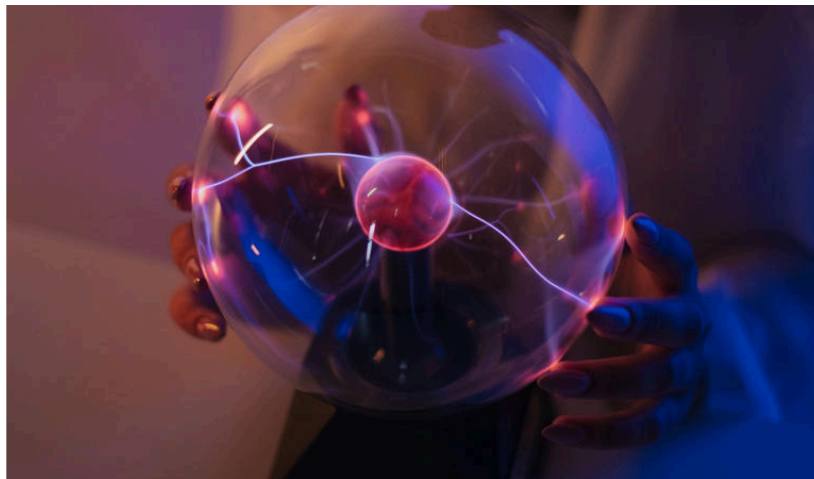
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A team of physicists at Oak Ridge National Laboratory has resolved a long-standing puzzle in quantum materials research: why do two honeycomb-lattice magnets with identical crystal structures behave so differently when cooled and reheated? Their answer — [published September 3, 2026](#), in *Physical Review Materials* and selected as an Editors' Suggestion — carries direct consequences

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for anyone building quantum computing or spintronic devices out of two-dimensional magnetic materials.



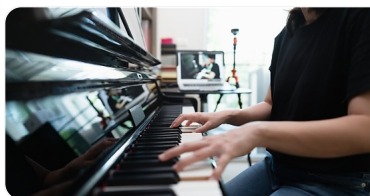
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Chromium trichloride (CrCl_3) and alpha-ruthenium trichloride ($\alpha\text{-RuCl}_3$) are, on a crystal-structure diagram, virtually indistinguishable: both belong to the layered transition-metal trihalide family, both host metal ions at the vertices of a honeycomb lattice, and both undergo the same type of temperature-driven structural rearrangement as they are cooled from room temperature. Yet in practice, CrCl_3 is structurally robust — it cycles through that rearrangement gracefully and recovers — while $\alpha\text{-RuCl}_3$ progressively degrades with each cycle, accumulating stacking faults that alter its magnetic ordering temperature, suppress its thermal conductivity, and make its experimental behavior nearly impossible to reproduce from laboratory to laboratory.

The study, led by Zachary Morgan of ORNL's Neutron Scattering Division together with Iris Ye, Jiasen Guo, and materials scientists Michael A. McGuire and Jiaqiang Yan, used neutron single-crystal diffraction at the Spallation Neutron Source to track both materials simultaneously through their structural transitions — the first direct, side-by-side comparison of this kind on single crystals of both compounds.



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Why the Crystal Matters: A Primer on Honeycomb Magnets

In a transition-metal trihalide, metal ions — chromium in CrCl_3 , ruthenium in $\alpha\text{-RuCl}_3$ — sit at the vertices of a honeycomb lattice inside each layer. Those layers are stacked on top of one another and held together not by strong chemical bonds but by weak van der Waals forces: the same gentle intermolecular attraction that allows graphite to flake off onto paper.

That weak interlayer coupling is what makes these materials so attractive for device engineering. It allows layers to be mechanically peeled away to atomically thin sheets, which can then be reassembled into heterostructures — layered sandwiches of different two-dimensional materials assembled like atomic-scale building blocks. Chromium trihalides including CrCl_3 have already been demonstrated as spin-filter magnetic tunnel junctions with [tunneling magnetoresistance values ranging](#) from 17,000% to 57,000%, opening a route toward voltage-controlled magnetic memory.



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But weak coupling also means the stacking arrangement is energetically fragile. At high temperature, both CrCl_3 and $\alpha\text{-RuCl}_3$ adopt a monoclinic structure in which adjacent layers are offset in a near-face-centered-cubic pattern. On cooling, both transition to a rhombohedral structure in which the layers adopt a different lateral offset — closer to hexagonal close-packing. This first-order structural phase transition, driven by the collective sliding of layers relative to one another, reshuffles the exchange pathways through which magnetic interactions are transmitted between layers.

The central question the ORNL team set out to answer: does that reshuffling happen the same way in both materials, and does it couple to magnetism the same way?

What ORNL's Neutrons Revealed

To answer that question, the team used neutron single-crystal diffraction — the technique of choice for probing both crystal structure and magnetic order simultaneously, because neutrons carry their own magnetic moment and scatter from both atomic and magnetic arrangements. By tracking Bragg peaks (sharp reflections from long-range crystalline order) and diffuse scattering (broader, smeared intensity from short-range correlations or disorder) as the temperature swept through the transition range, they could watch each material rearrange and document what happened on either side of the transition.

The contrast was stark.

CrCl_3 handled the transition smoothly. Its in-plane lattice parameter — the spacing between chromium ions within a single honeycomb plane — evolved continuously and without any abrupt jump as the temperature passed through the transition. The structural changes were reversible: heat the crystal through the transition, cool it back down, and it returned to essentially the same state it started from, with no progressive accumulation of damage across repeated cycles.

Magnetically, CrCl_3 ordered into an A-type antiferromagnetic structure at a Néel temperature of 14 K — a configuration in which adjacent honeycomb planes align their spins antiparallel, while spins within each plane align ferromagnetically. But what stood out was what happened above that ordering temperature: diffuse magnetic scattering — the neutron signature of short-range spin correlations — persisted [all the way up to roughly 40 K](#), some 26 degrees above the point where true long-range magnetic order snaps into place. CrCl_3 's spins remain correlated and locally organized long before they commit to global magnetic order.

$\alpha\text{-RuCl}_3$ told the opposite story. Its in-plane lattice parameter did not evolve smoothly — it jumped discontinuously across the structural transition, showing pronounced hysteresis between the heating and cooling legs of the same thermal cycle. More critically, repeating the cycle caused the crystal to progressively degrade. Each thermal trip accumulated stacking faults that did not heal on cooling — a ratchet-like buildup of crystalline disorder.

Magnetically, α - RuCl_3 showed a complete absence of diffuse scattering above its ordering temperature of 7.6 K. Where CrCl_3 's spins remained correlated far above T_N , those in α - RuCl_3 appeared to disengage abruptly at the ordering temperature, leaving no short-range magnetic precursor.

What Is the Structural Transition, and Why Does It Differ?

The key to understanding the contrast lies in the fundamentally different electronic characters of chromium and ruthenium in these compounds.

Chromium in CrCl_3 is a 3d transition metal — a lighter element — with three electrons in its d-shell, all with parallel spins. Its magnetic interactions are dominated by conventional Heisenberg-type exchange: isotropic, relatively short-ranged, and insensitive to the precise geometry of the crystal lattice beyond nearest-neighbor distances. When CrCl_3 layers slide into a new stacking configuration during the structural transition, the magnetic exchange pathways shift, but the Heisenberg exchange adapts smoothly. The in-plane bonds do not need to strain abruptly to accommodate the new configuration.

Ruthenium in α - RuCl_3 is a 4d metal — heavier — and its greater atomic number brings [strong spin-orbit coupling](#) into play. Spin-orbit coupling is the relativistic interaction between an electron's spin and its orbital motion; in heavier elements like ruthenium, it is strong enough to fundamentally entangle spin and orbital degrees of freedom into a new effective quantum number ($l_{\text{eff}} = 1/2$)



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instead hosting exotic fractionalized excitations called Majorana fermions. The [2019 review of Kitaev materials](#) by Takagi, Takayama, Jackeli, Khaliullin, and Nagler provides a comprehensive account of the concept and its material realizations.

The magnetic interactions that produce this state — the Kitaev couplings — are bond-dependent and highly anisotropic: each of the three types of bonds in the honeycomb lattice carries a different spin component interaction. This means the interactions are exquisitely sensitive to the precise geometry of the ruthenium-chloride-ruthenium exchange pathways — and hence to the exact layer stacking. When α - RuCl_3 layers slide during the structural transition, the resulting change in exchange pathway geometry is not something the in-plane lattice can accommodate elastically. The abrupt jump in the in-plane lattice parameter — and the progressive damage — is the mechanical consequence of that geometric sensitivity.

In CrCl_3 , by contrast, the Heisenberg-dominated exchange imposes no equivalent geometric constraint. Layers slide, the stacking rearranges, and the in-plane bonds remain unperturbed.

Why This Is Costing Quantum Computing Researchers Sleep

The structural fragility of α - RuCl_3 is not merely an inconvenience for experimentalists. It is the root cause of one of the most consequential reproducibility problems in contemporary condensed matter physics.

For the past decade, α -RuCl₃ has been the community's best bet for observing Majorana fermions in a material setting — not as individual exotic particles in expensive cryogenic setups, but as collective excitations in a relatively ordinary-looking black crystal. Topological quantum computers would use Majorana fermions to store information in a form that is inherently resistant to decoherence: rather than isolating a qubit from its environment (the conventional approach), topological quantum computing would encode information in the global entanglement pattern of the spin liquid itself, making it robust to any local perturbation.

The smoking-gun signature of that state is a half-quantized thermal Hall conductance — a precise fractional value that emerges from Majorana edge modes when a magnetic field is applied to α -RuCl₃. Several research groups have [reported seeing this half-quantized value](#). Several others have not been able to reproduce it, or have measured values that fall far short of quantization — a controversy [documented in preprint form](#) and debated actively across the field.

Prior ORNL work published in *Physical Review Materials* in 2024 — by McGuire, Yan, and co-workers, several of whom are also authors on the current study — showed that [the thermal Hall conductance measurement](#) in α -RuCl₃ is strongly sample-dependent, and that samples with more stacking disorder systematically show lower values that push the measured result further from the half-quantized target. Crystals with minimal stacking disorder show higher thermal conductivity and a thermal Hall conductance that approaches the predicted quantized value.

The current study explains *why* stacking disorder accumulates in α -RuCl₃: the abrupt in-plane lattice jump during the first-order structural transition generates elastic strains that damage the crystal's layer stacking — and that damage adds up with every thermal cycle a sample undergoes during growth, post-growth annealing, or repeated experimental cycling. CrCl₃ does not have this problem because its Heisenberg-dominated exchange does not lock the in-plane bonds to the stacking geometry the way Kitaev coupling does in α -RuCl₃.

In other words: the property that makes α -RuCl₃ scientifically extraordinary — the bond-directional, spin-orbit-driven Kitaev exchange — is the same property that makes its crystals mechanically fragile. The exotic physics and the structural weakness share a common root.

What Are the Structural Transitions, Exactly?

Both CrCl₃ and α -RuCl₃ switch between two well-defined crystal structures as they pass through the structural transition. At high temperature, both adopt a monoclinic arrangement (space group C2/m), in which adjacent honeycomb layers are offset from one another by a lateral displacement that corresponds roughly to face-centered-cubic layer ordering. On cooling below the transition temperature — which occurs near 240 K (–33°F) for CrCl₃ and near 150 K (–189°F) for α -RuCl₃ — both switch to a rhombohedral arrangement (space group R $\bar{3}$), in which the lateral displacement corresponds to a different close-packed ordering.

The difference the ORNL team documented is in how that switching happens in the two materials. In CrCl₃, the lattice breathes smoothly through the change. In α -RuCl₃, the abruptness of the transition — visible as a discontinuous jump in the in-plane lattice parameter, and as a pronounced difference between the heating and cooling curves — is the mechanical signature of a system whose in-plane bonding chemistry is being pulled into conflict with its stacking geometry as the layers try to slide.

The c-lattice parameter (the spacing between layers, measured perpendicular to the honeycomb planes) shows a step-like change in both materials — consistent with the expected stacking rearrangement. It is the in-plane response that diverges, and that divergence is the key finding.

What This Tells Device Engineers

For researchers building spintronic devices from van der Waals magnetic layers, the study's message is straightforward: structural reversibility under thermal cycling cannot be assumed from crystal-structure analogy alone.

CrCl_3 and its chromium trihalide relatives — CrBr_3 and CrI_3 — have attracted intense interest as magnetic layers in van der Waals heterostructure devices. In those architectures, a magnetic CrX_3 layer sandwiched between graphene contacts creates an atomically thin spin filter: switching the magnetization state of the CrX_3 layer changes the tunneling resistance dramatically, enabling voltage-controlled magnetic memory and logic without traditional magnetic materials. The ORNL result validates CrCl_3 's suitability for those applications by confirming that it cycles through its structural transition cleanly and without accumulating disorder that would unpredictably shift its magnetic switching behavior. Research on [CrCl₃ flakes to monolayer thickness](#) further supports the path toward scalable device manufacturing.

For $\alpha\text{-RuCl}_3$ and topological quantum computing, the implications are more sobering in the short term. The progressive stacking disorder accumulation means that samples degrade with use, making sample selection, preparation, and characterization not incidental to the experiment but central to it. The ORNL team's prior work established criteria for selecting minimal-disorder crystals: a structural transition near 140 K on cooling, and a single magnetic anomaly near 7.6 K in specific heat. The new study explains the mechanism behind those criteria — a crystal that shows the transition cleanly near 140 K has not yet accumulated the in-plane strain damage that pushes the transition temperature down and smears it.

More broadly, the study establishes that in van der Waals magnets, structural analogy does not imply functional equivalence: materials with the same space group and the same nominal crystal architecture can differ fundamentally in how their layers respond to thermal stress, and how that stress couples back to magnetic behavior. For quantum device engineering, where reproducibility is not a statistical luxury but an operational requirement, the coupling between structural mechanics and electronic configuration is a design variable that must be tested explicitly.

How Neutron Diffraction Made This Visible

The methodological choice at the heart of the study — neutron single-crystal diffraction using the [CORELLI elastic diffuse scattering spectrometer](#) at ORNL's Spallation Neutron Source — was critical to the findings. Neutrons interact with both the nuclear and magnetic structure of a material in a single experiment, allowing the team to track structural and magnetic order simultaneously as a function of temperature, rather than running separate measurements that might reflect slightly different sample conditions.

The ability to distinguish Bragg peaks (which carry information about the long-range crystalline order) from diffuse scattering (which reveals departures from that order — stacking faults, short-range correlations, and disordered domains)

gave the team direct access to the disorder that is invisible to conventional X-ray diffraction but critical to the physical properties being debated.

Frequently Asked Questions

What is a Kitaev quantum spin liquid, and why does it matter for quantum computing?

A Kitaev quantum spin liquid is a state of matter predicted by mathematician Alexei Kitaev in 2006, in which spins on a honeycomb lattice are quantum mechanically entangled in a specific topological pattern without ever settling into ordinary magnetic order. The excitations of this state — called Majorana fermions — obey a type of quantum statistics (non-Abelian anyons under a magnetic field) that could be used to store and process quantum information in a form that is inherently resistant to the environmental disturbances that cause ordinary qubits to fail. Alpha-ruthenium trichloride (α -RuCl₃) is currently the most studied material candidate for hosting this state.

Why has α -RuCl₃ been so difficult to study reproducibly across different laboratories?

Multiple research groups have reported conflicting measurements of the same key property — the thermal Hall conductance, which should show a precise half-integer quantized value if Majorana fermions are present. Prior ORNL work established that the measurement is strongly sample-dependent: crystals with more stacking disorder (layers offset from their ideal positions) produce lower thermal conductivity and measurements that fall further from the predicted quantized value. The new Morgan et al. study explains why stacking disorder accumulates in α -RuCl₃: the abrupt, hysteretic jump in the in-plane lattice parameter during the structural transition generates elastic damage that adds up with each thermal cycle. This structural explanation for the reproducibility problem is a significant advance.

What makes CrCl₃ a better candidate for van der Waals spintronic devices than α -RuCl₃?

CrCl₃'s magnetism is governed by isotropic Heisenberg exchange, which is not geometrically sensitive to the precise stacking arrangement of the honeycomb layers. As a result, CrCl₃ cycles through its structural phase transition smoothly — the in-plane lattice parameter evolves continuously, with no abrupt jump, and the crystal does not accumulate damage. This structural reversibility means CrCl₃ can withstand the thermal cycling inherent in device manufacturing and repeated operation without progressively shifting its magnetic properties — a requirement that α -RuCl₃, which degrades with each cycle, cannot currently meet. [Research on growing CrCl₃ to monolayer thickness](#) confirms its promise for scalable spintronic applications.

What is diffuse magnetic scattering, and what does the difference between the two materials

tell us?

Diffuse magnetic scattering is the broad, spread-out neutron signal that arises when spins in a material are correlated over short distances — organizing themselves into local patches of order — without yet committing to the long-range, fully-ordered magnetic state. In CrCl_3 , this diffuse scattering persists up to about 40 K, roughly 26 K above the temperature at which true long-range magnetic order locks in at 14 K. That persistence reflects the gradual, isotropic nature of Heisenberg exchange: spins build up local coherence progressively over a wide temperature range before snapping into global order. In $\alpha\text{-RuCl}_3$, no equivalent diffuse scattering was detected above the 7.6 K ordering temperature — consistent with the anisotropic, bond-directional Kitaev exchange, which suppresses the conventional precursor fluctuations that CrCl_3 's Heisenberg exchange readily supports.

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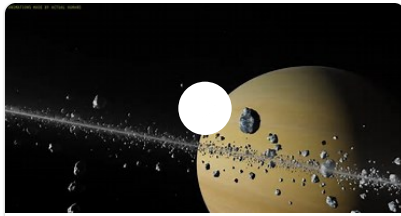
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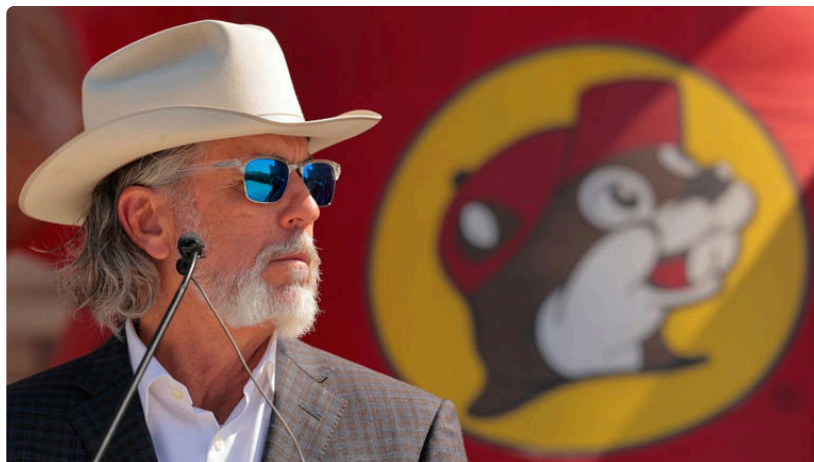
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Buc-ee's CEO said company is focused on 'conservative, business-friendly' states

Story by Joey Esposito · 5d · 3 min read



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Claim:

In August 2026, Buc-ee's convenience store chain CEO Arch "Beaver" Aplin III said the company was focused on building new stores in "conservative, business-friendly" states.

Rating:

Correct Attribution ([About this rating?](#))

A rumor that the CEO of a popular chain of convenience stores in the southern United States said the the company would no longer pursue locations in so-called "blue states" circulated online in August 2026.

Users on [Facebook](#) and a popular thread on [Reddit](#) suggested Buc-ee's CEO Arch "Beaver" Aplin III said the company was focused on building in "conservative, business-friendly states."



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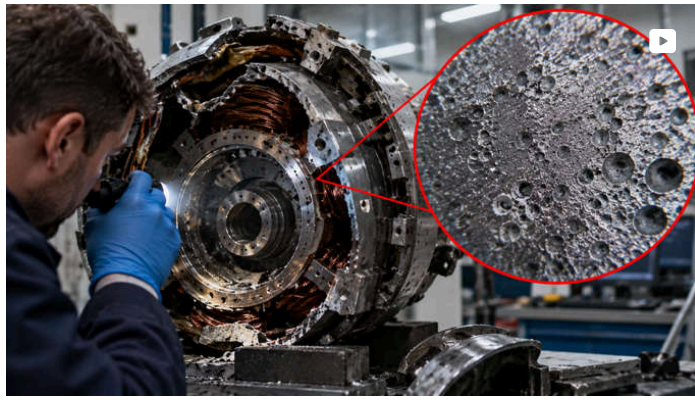
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